

Demand Forecasting & Inventory Optimization

B.Tech Project Technical Report - Supply Chain Analytics

1. Data Acquisition and Characteristics

Concept: Demand in supply chains is a composite of a base level, long-term trends, and seasonal cycles. Real-world data often contains noise and holiday-related spikes.

Dataset generated for 48 months (2021-2024).

Initial Demand Level: 300 units.

Growth Trend: 5 units/month.

2. Exploratory Data Analysis (EDA)

Concept: Moving averages act as a low-pass filter to remove short-term noise and reveal the underlying direction of the supply chain. Boxplots identify periodic seasonal variance.

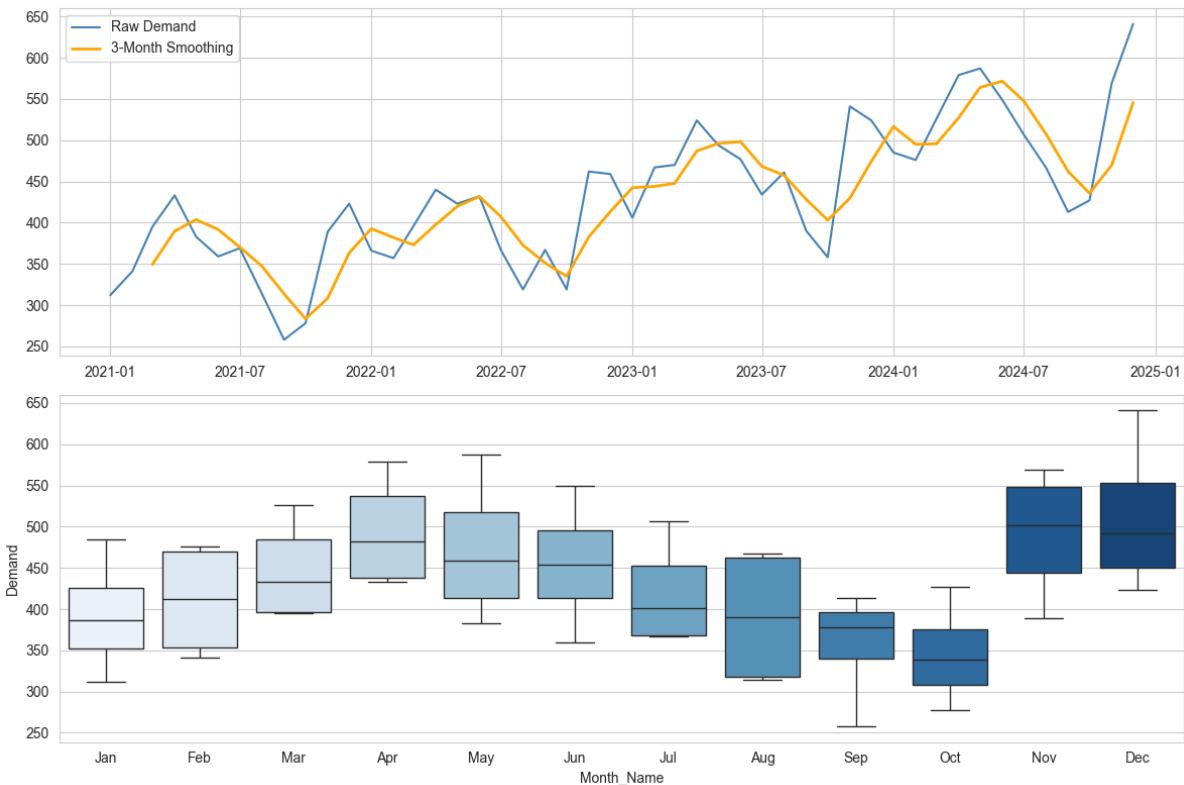


Figure: Demand Trends and Monthly Distribution Analysis

3. Time Series Decomposition

Concept: Decomposition deconstructs demand into Trend, Seasonal, and Residual components. This allows managers to distinguish between permanent growth and temporary seasonal spikes.

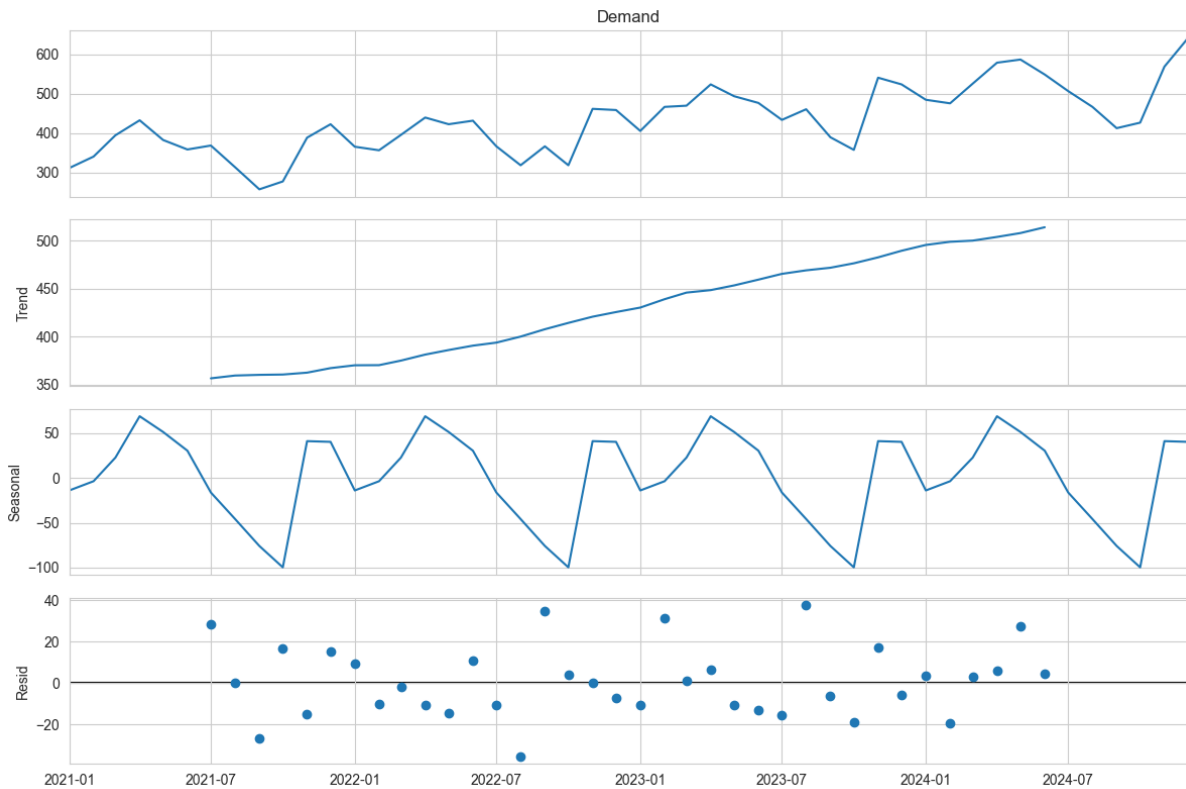


Figure: Additive Decomposition showing isolated Trend and Seasonality

4. Statistical Stationarity Testing

Concept: Stationarity implies that mean and variance are constant over time. If a series is non-stationary, models like SARIMA cannot be reliably applied without differencing.

Augmented Dickey-Fuller Test Results:

ADF Statistic: 0.3148

p-value: 0.9780

Conclusion: Non-Stationary

5. Correlation Analysis (ACF & PACF)

Concept: Autocorrelation (ACF) identifies the 'memory' of the demand, while Partial Autocorrelation (PACF) determines the direct relationship between lags, used to set SARIMA orders (p , q).

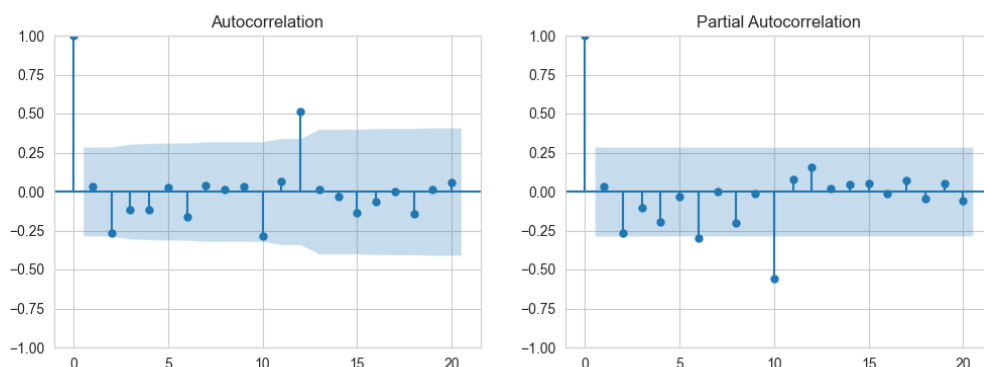


Figure: ACF and PACF Plots of Differenced Demand Data

6. Forecasting Model Performance

Concept: We compare Holt-Winters (Exponential Smoothing) against SARIMA (Stochastic Modeling). The Mean Absolute Percentage Error (MAPE) serves as the primary accuracy metric.

Model Name	MAE	MAPE (%)
Holt-Winters	23.2	4.66 %
SARIMA	30.9	5.91 %

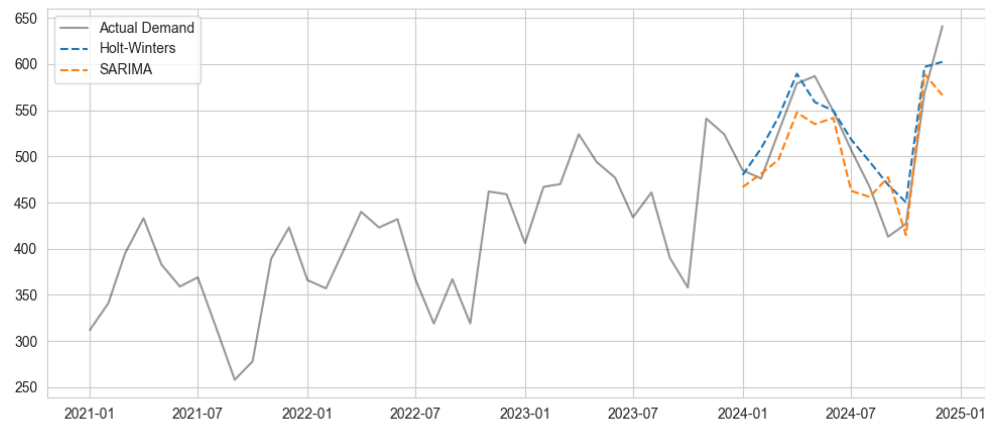


Figure: Comparative Forecast Visualization

7. Inventory Strategy Optimization

Concept: Safety Stock provides a buffer against forecast error. EOQ balances ordering costs against holding costs. The Reorder Point (ROP) triggers replenishment based on lead time demand.

Safety Stock: 167 units
Reorder Point (ROP): 697 units
Economic Order Quantity (EOQ): 797 units



Figure: Inventory Level Simulation over Time